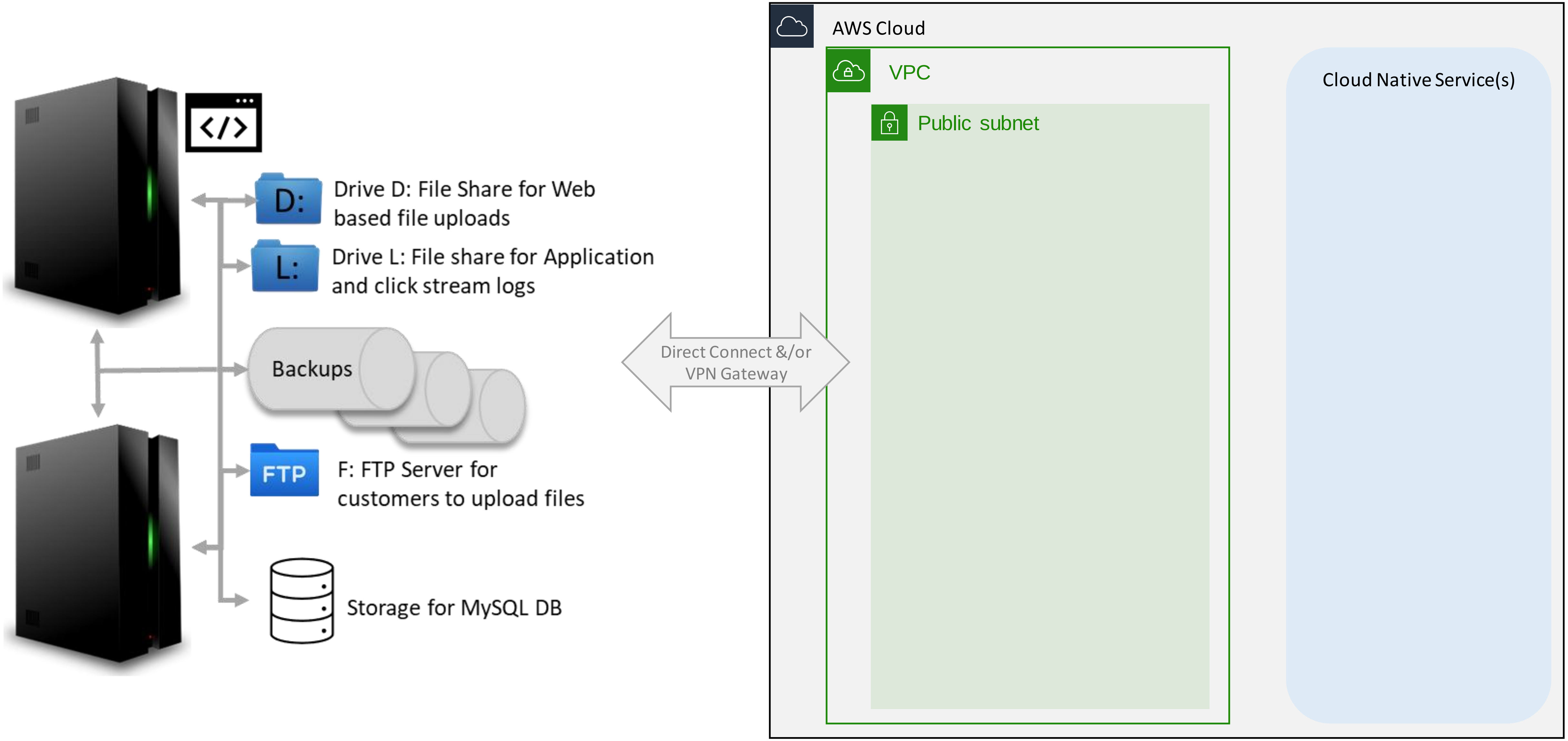
Migrating Application to AWS

Use case



Drive D: File Share for Web based file uploads
Drive L: File share for Application and click stream logs
Backups
F: FTP Server for customers to upload files
Storage for MySQL DB
D:
L:
FTP
Direct Connect &/or VPN Gateway
AWS Cloud
VPC
Public subnet
Cloud Native Service(s)

Migrated Version – Representative Architecture

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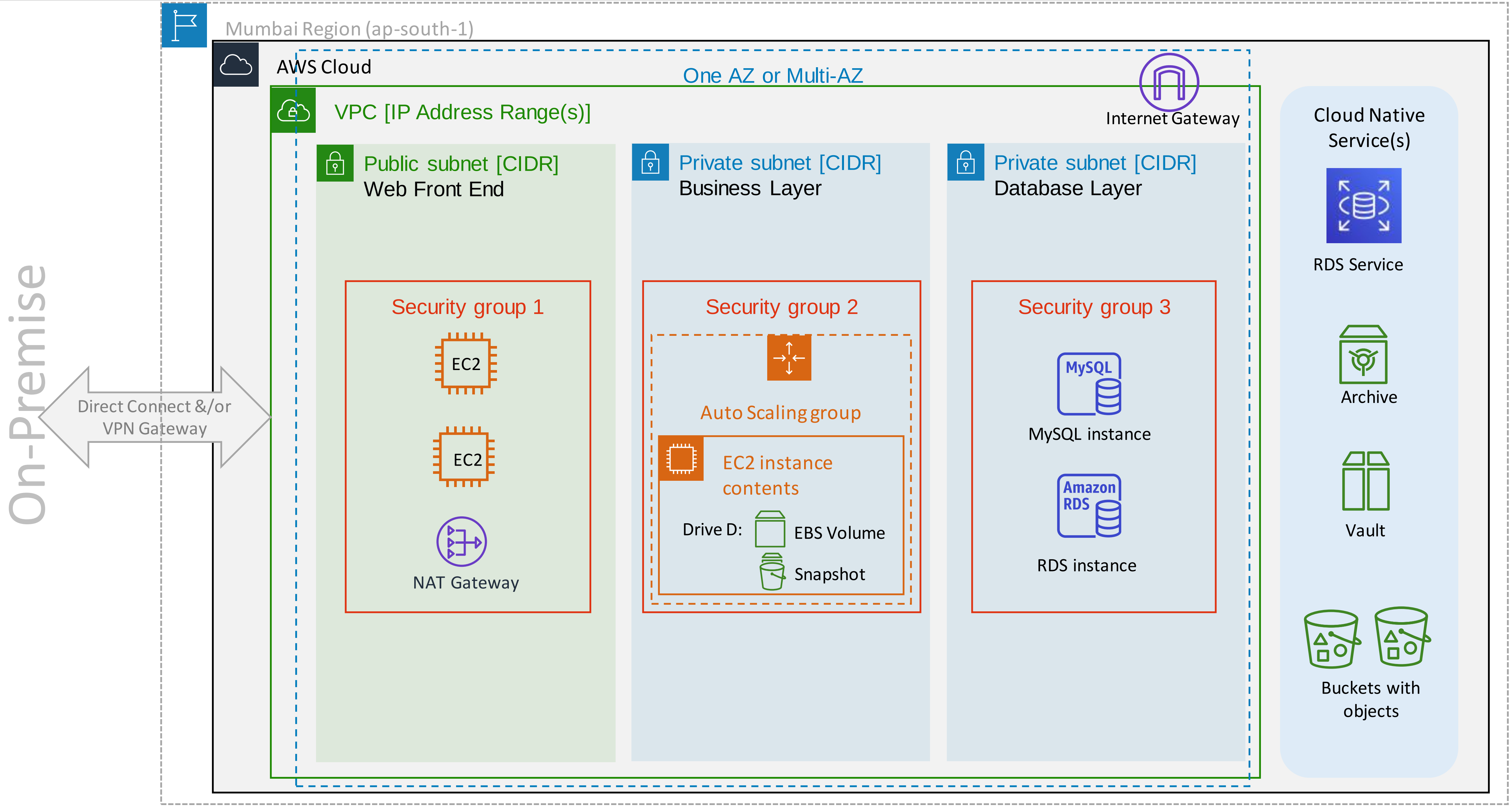
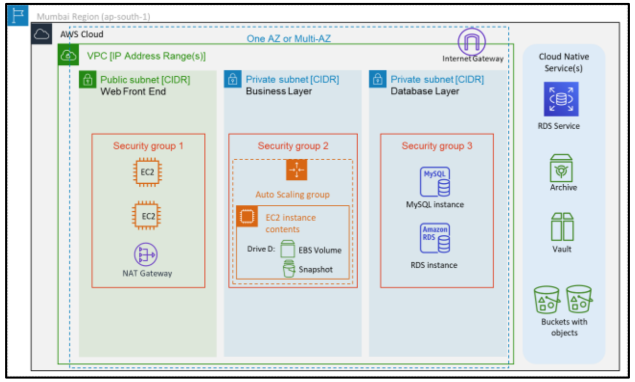


Post-Migration Activities

North Side
Operations
Monitoring
Administrators

West Side
Engineering Team
Architectures
QA Team
Program Managers
Project Managers



East Side
End Users
B2B Interface
B2C Interface

Malicious users

South Side
Auditors
Compliance
Rules & Regulations

AWS Security

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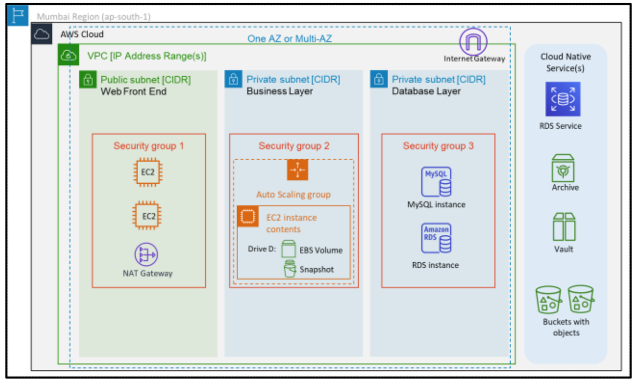

[WAF](https://aws.amazon.com/waf/)

End Users
B2B Interface
B2C Interface

Malicious users

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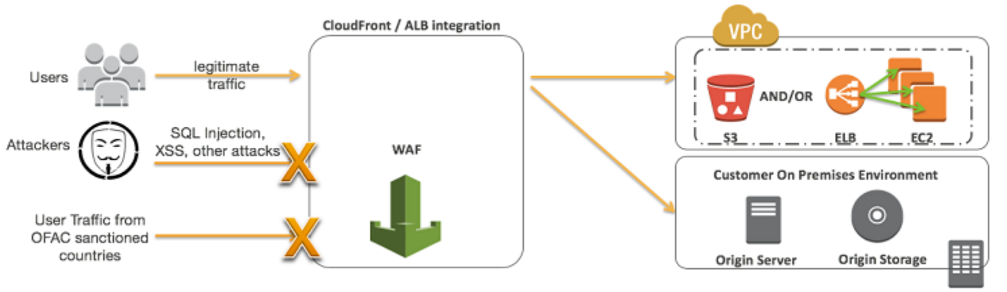

https://www.ofac-guide.com/

Features	Description
Web Traffic Filtering	AWS WAF lets you create rules to filter web traffic based on conditions that include IP addresses, HTTP headers and body, or custom URIs.
Full feature API	AWS WAF can also be deployed and provisioned automatically with AWS CloudFormation sample templates that allow you to describe all security rules you would like to deploy for your web applications delivered by Amazon CloudFront
Real-time visibility	AWS WAF provides real-time metrics and captures raw requests that include details about IP addresses, geo locations, URIs, User-Agent and Referrer's. AWS WAF is fully integrated with Amazon CloudWatch.

AWS Security

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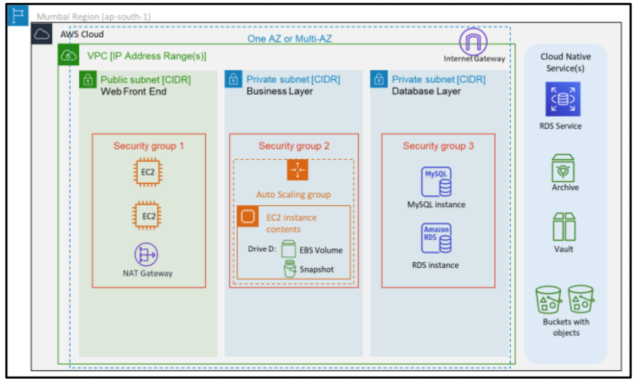

[WAF](#)

[Shield](#)

End Users

B2C Interface

Malicious users

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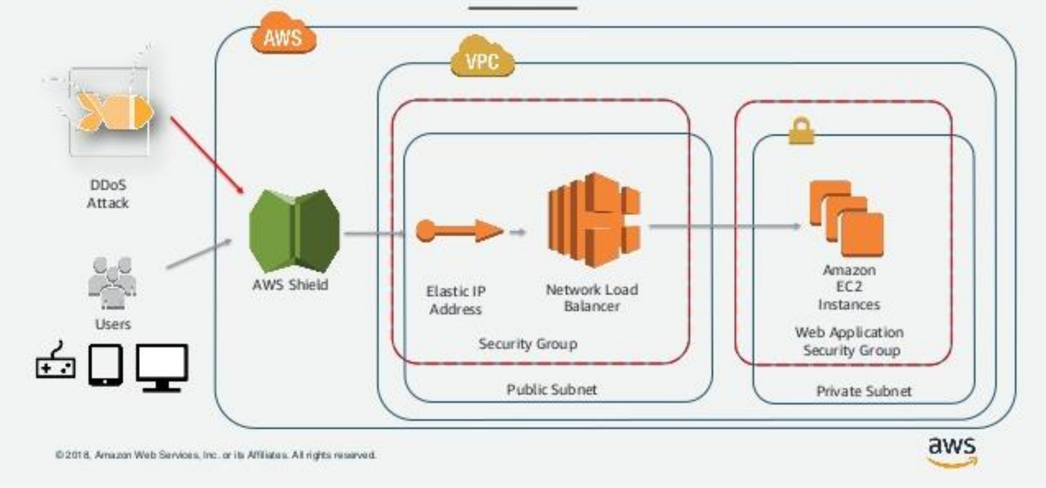

Features	Description
Quick detection	AWS Shield Standard provides always-on network flow monitoring which inspects incoming traffic to AWS and uses a combination of traffic signatures, anomaly algorithms and other analysis techniques to detect malicious traffic in real-time
Inline attack mitigation	Automated mitigation techniques are built-into AWS Shield Standard, giving you protection against common, most frequently occurring infrastructure attacks. When you use AWS Shield Standard with Amazon CloudFront and Amazon Route 53, you receive comprehensive availability protection against all known infrastructure (Layer 3 and 4) attacks

[See live threat detections here](#)

AWS Security

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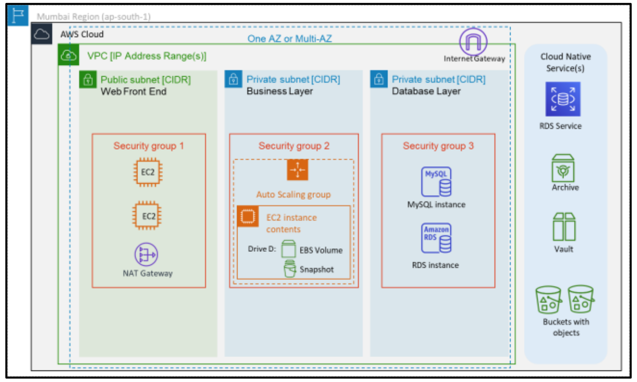

[WAF]
[Shield]
[GuardDuty]

Malicious users

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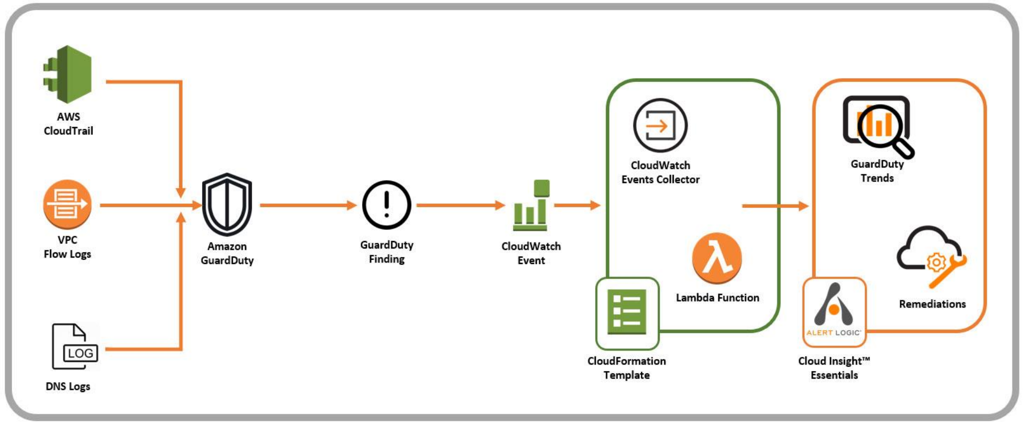

Description

A threat detection service that continuously monitors for malicious activity and unauthorized behavior to protect your AWS accounts and workloads.

An intelligent and cost-effective option for continuous threat detection in the AWS Cloud.

Service uses machine learning, anomaly detection, and integrated threat intelligence to identify and prioritize potential threats.

Analyzes tens of billions of events across multiple AWS data sources, such as AWS CloudTrail, Amazon VPC Flow Logs, and DNS logs.

Integration with AWS CloudWatch Events, GuardDuty alerts are actionable, easy to aggregate across multiple accounts, and straightforward to push into existing event management and workflow systems.

AWS Auditing & Compliance

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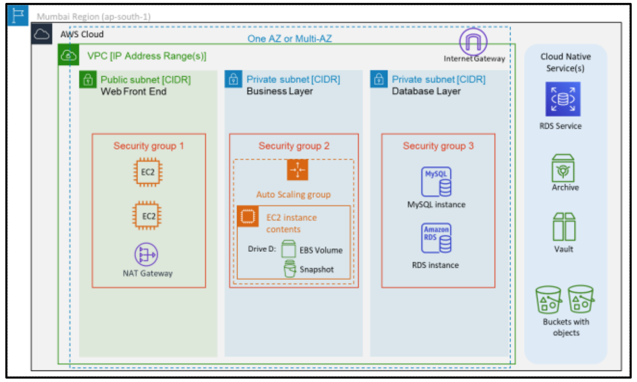


WAF

End Users

Shield

B2C Interface

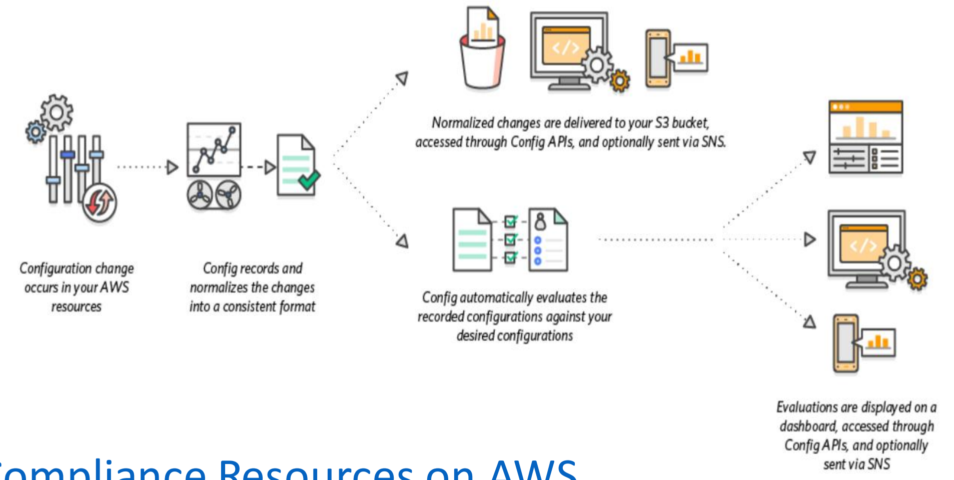
GuardDuty

Malicious users

So

Config

Auditors
Compliance
Rules & Regulations



[Compliance Resources on AWS]()

Description

Description
Continuously monitor, Record configuration changes & Inventory your AWS resources
Allows you to continuously audit and assess the overall compliance of your AWS resource configurations with your organization’s policies and guidelines
Track the relationships among resources and review resource dependencies prior to making changes. (as CMDB)
Helps identify the root cause of operational issues through its integration with AWS CloudTrail
With multi-account, multi-region data aggregation in AWS Config, you can view compliance status across your enterprise and identify non-compliant accounts from one central console

AWS Auditing & Compliance

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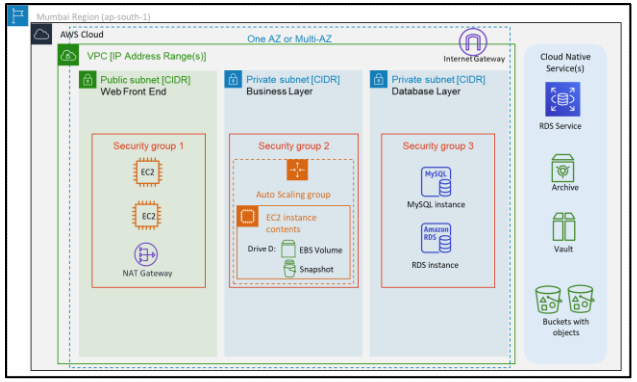


WAF

Shield

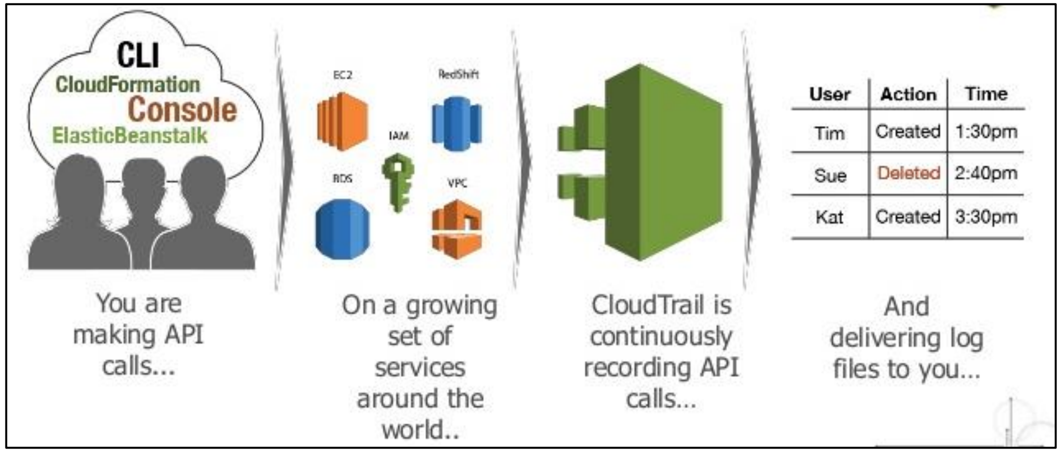
GuardDuty

Malicious users

Config

CloudTrail

Rules & Regulations



Description

Service that enables governance, compliance, operational auditing, and risk auditing of your AWS account.

Provides event history of your AWS account activity, including actions taken through the AWS Management Console, AWS SDKs, command line tools, and other AWS services

Event history simplifies security analysis, resource change tracking, and troubleshooting edureka!

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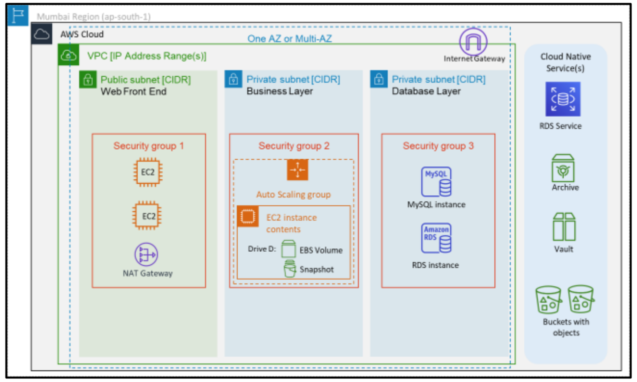


[WAF](#)

[Shield](#)

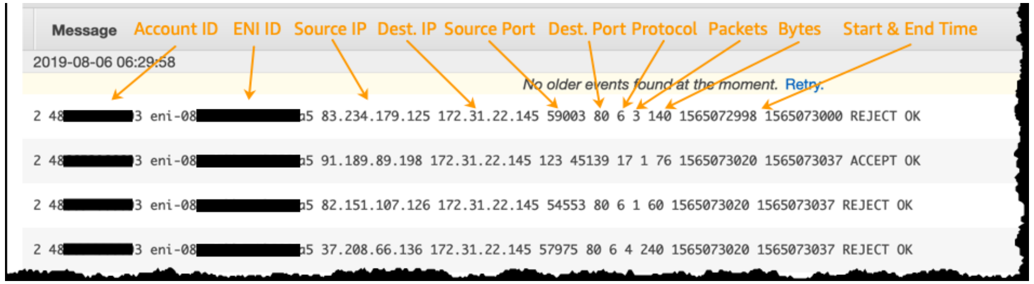
[GuardDuty](#)

Malicious users

[Config](#)

[CloudTrail](#)

[VPC Flow Log](#)



Description

Enables you to capture information about the IP traffic going to and from network interfaces in your VPC. Can be published to CloudWatch Logs and S3 for analysis.

Helps in

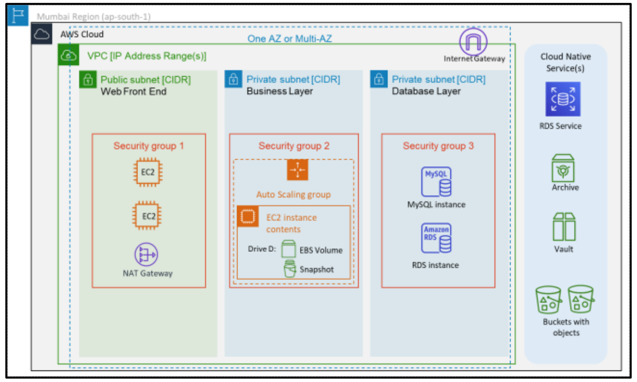
- Diagnosing overly restrictive security group rules
- Monitoring the traffic that is reaching your instance
- Determining the direction of the traffic to and from the network interfaces

Collected outside of the path of your network traffic, and therefore does not affect network throughput or latency.

AWS Monitoring and Visualization

CloudFormation
Create scripts for solution

CloudFormation
Execute scripts



WAF

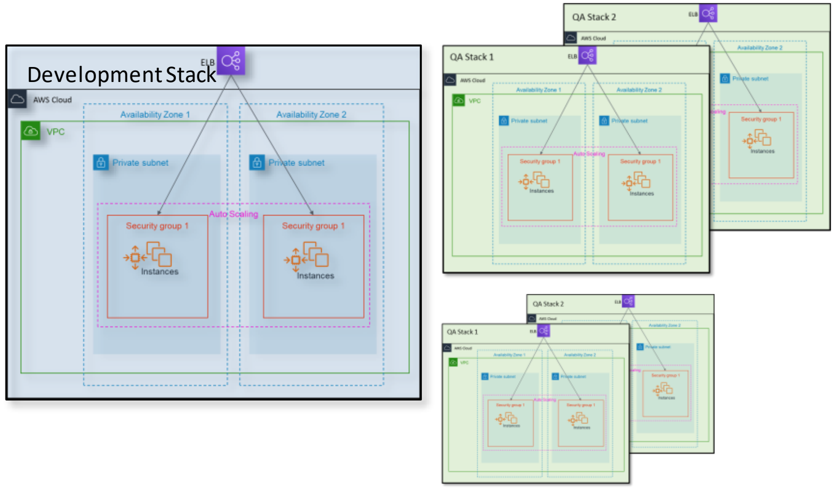
Shield

GuardDuty

Config

CloudTrail

VPC Flow Log



- Infrastructure as Code (IaC)
- AWS CDK
  - Cloud Development Kit
- Ready Templates available
- Designer Facility
- Parameterized
- When to use
  - Replicate complete infrastructure consistently for QA, UAT, Pre-Prod and Production
  - Saves time on building similar infrastructure
  - Manage dependencies
  - Preview changes done to infrastructure

AWS Monitoring and Visualization

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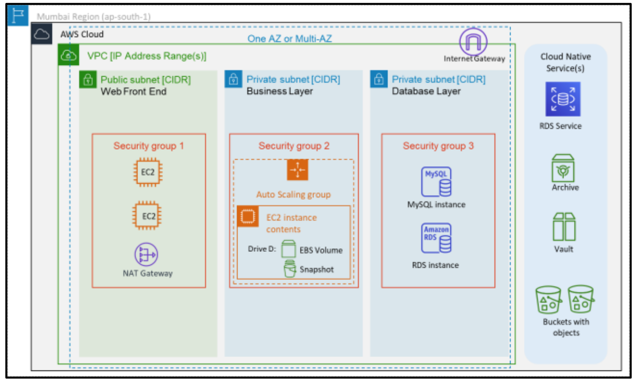
[CloudWatch](#)

[CloudFormation](#)
Execute scripts

[CloudFormation](#)
Create scripts for solution



[WAF](#)

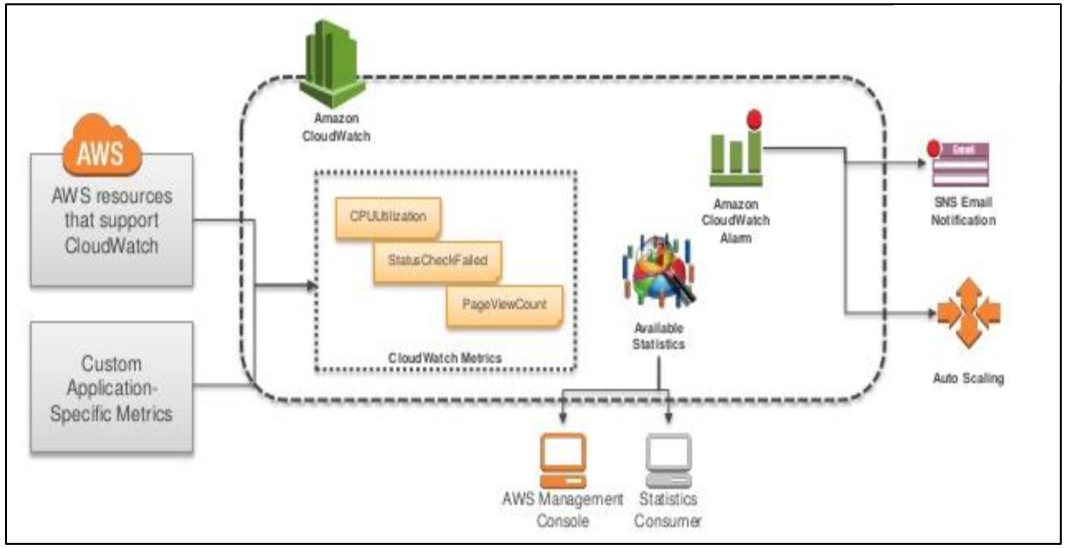
[Shield](#)

[GuardDuty](#)

[Config](#)

[CloudTrail](#)

[VPC Flow Log](#)



Description

Monitoring and observability service built for engineers, developers, site reliability engineers (SREs), and IT managers.

Collects and stores data (in the form of logs, metrics, and events) for actionable insights to monitor your applications, respond to system-wide performance changes, optimize resource utilization, and get a unified view of operational health for cloud based and on-premise servers.

Detects anomalous behavior in your environments, set alarms, visualize logs and metrics side by side, take automated actions, troubleshoot issues, and discover insights to keep your applications running smoothly

[Stop & Start Amazon EC2 instances at regular intervals using Lambda](#)

[Sample Query: fields @timestamp, @message | sort @timestamp desc | limit 25](#)

AWS Simple Systems Manager (SSM)

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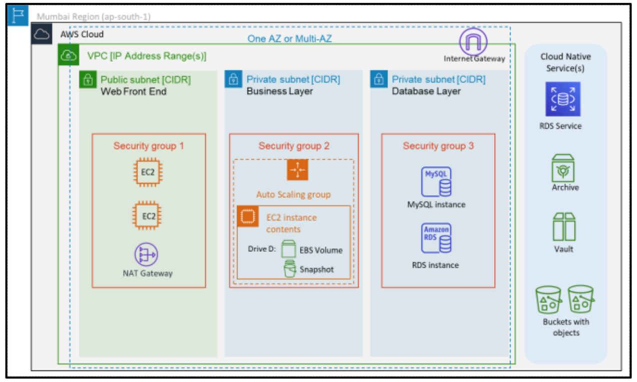
Systems Management

CloudWatch

CloudFormation
Execute scripts

CloudFormation
Create scripts for solution



WAF

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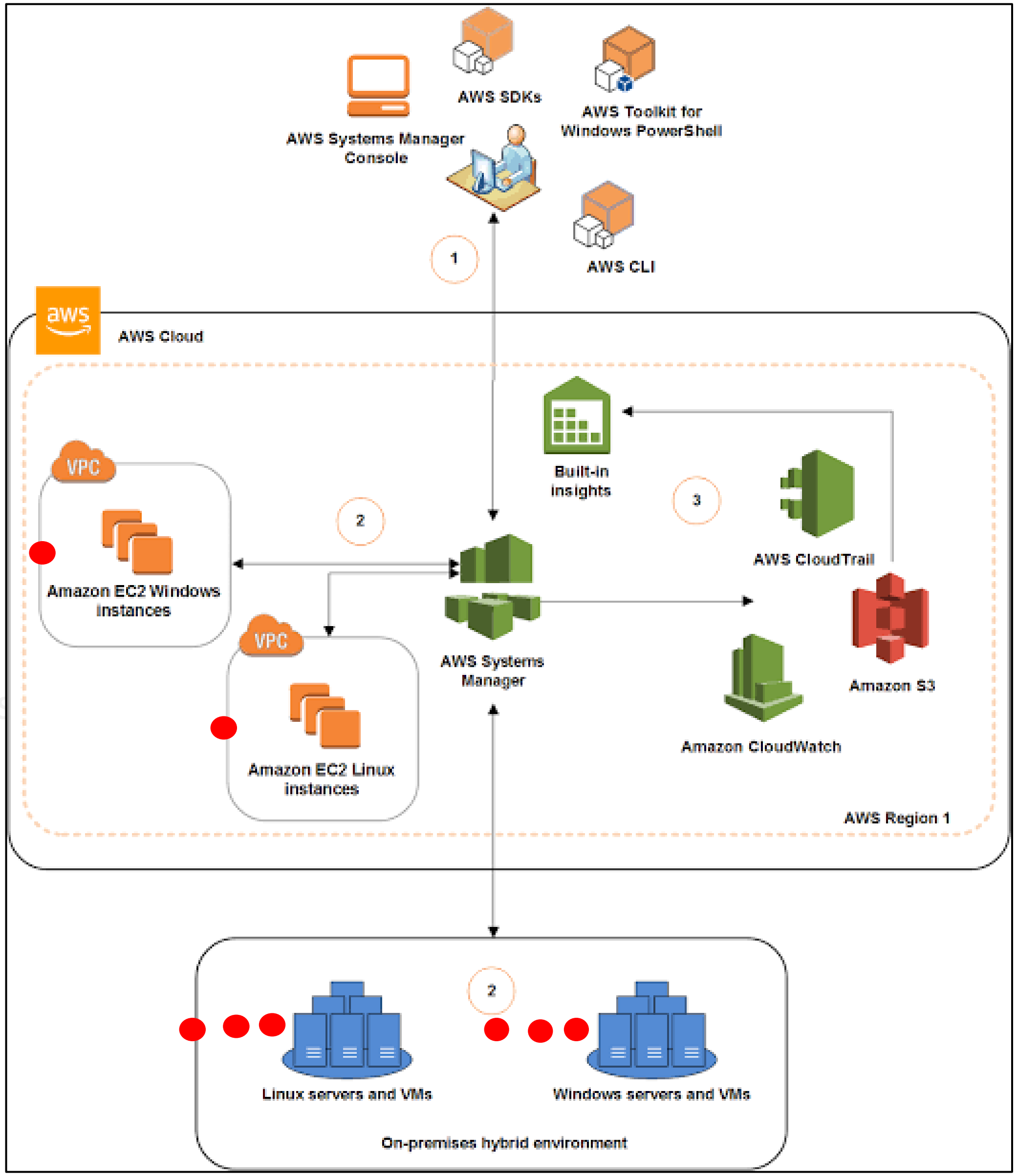
Config

CloudTrail

VPC Flow Log



SSM Agent

Automation with AWS CodeDeploy

AWS Backup

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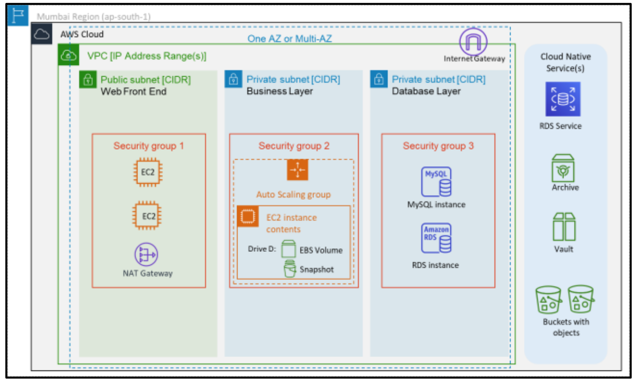
Backup

Systems Management

CloudWatch

CloudFormation
Execute scripts

CloudFormation
Create scripts for solution



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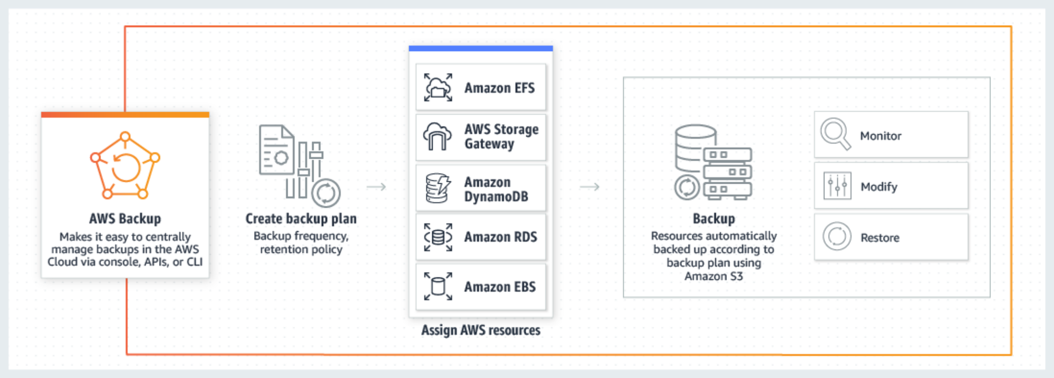

- Centralized backup management
- Policy-based backup solution
- Tag-based backup policies
- Automated backup scheduling
- Automated retention management
- Backup activity monitoring
- Lifecycle management policies
- Incremental backups
- Backup data encryption
- Backup access policies
- Amazon EC2 instance backups
- Item-level recovery for Amazon EFS
- Cross-region backup

Other common services for Engineering & Operations

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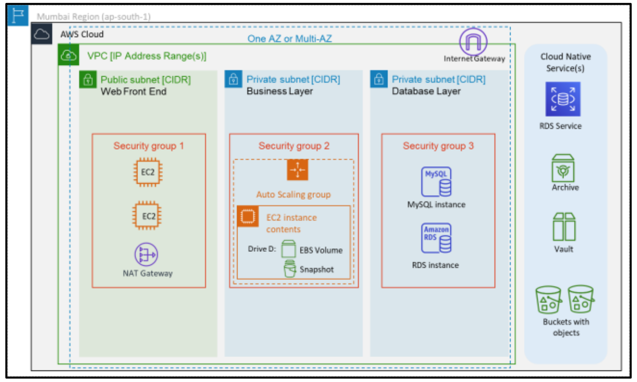
Backup

Systems Management

CloudWatch

CloudFormation
Execute scripts

CloudFormation
Create scripts for solution



WAF

Shield

GuardDuty

Management Console

OpsWorks

Personal Health Dashboard

Config

CloudTrail

VPC Flow Log

Container Technology

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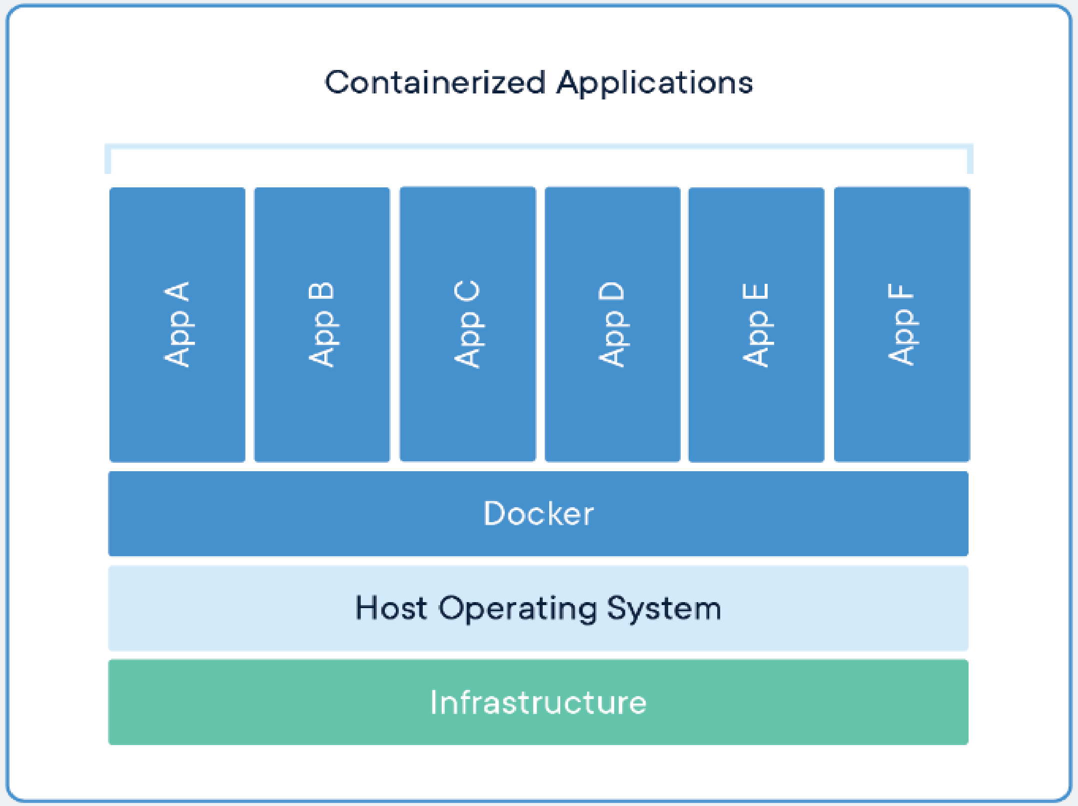
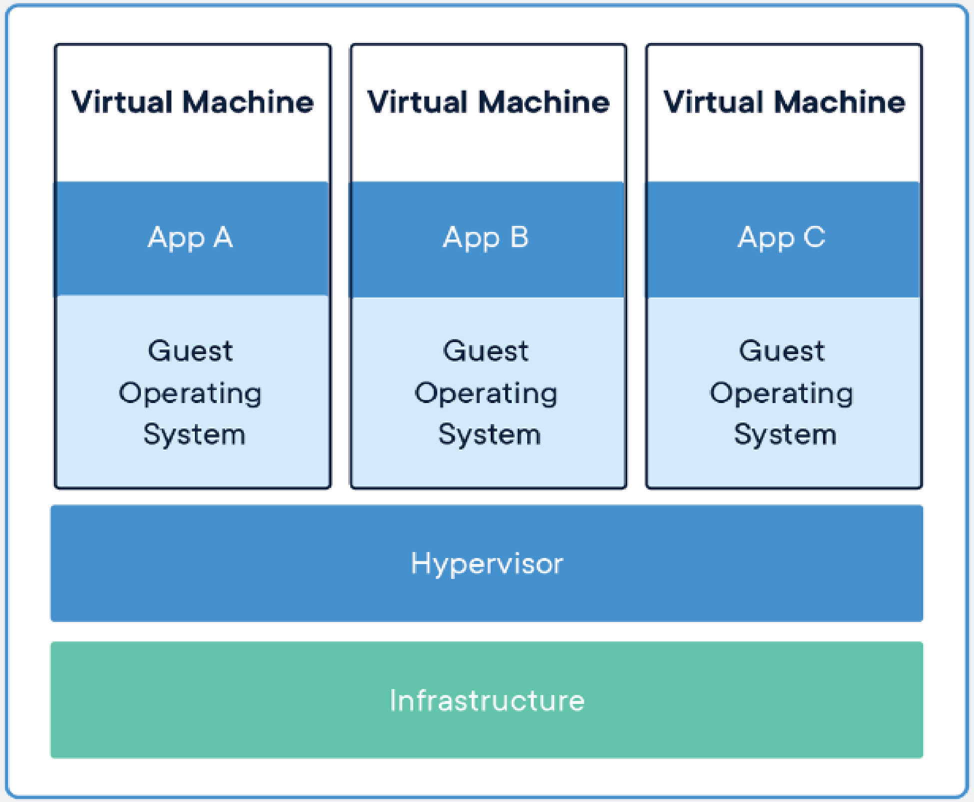
What is a container?





CONTAINERS

Containers are an abstraction at the app layer that packages code and dependencies together. Multiple containers can run on the same machine and share the OS kernel with other containers, each running as isolated processes in user space. Containers take up less space than VMs (container images are typically tens of MBs in size), can handle more applications and require fewer VMs and Operating systems.

VIRTUAL MACHINES

Virtual machines (VMs) are an abstraction of physical hardware turning one server into many servers. The hypervisor allows multiple VMs to run on a single machine. Each VM includes a full copy of an operating system, the application, necessary binaries and libraries - taking up tens of GBs. VMs can also be slow to boot.

Docker Basics

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#	Docker is NOT a
1	LXC – LinuxContainer
2	Virtual Machine Solution
3	Configuration Management System
4	Replacement for Chef, puppet, Ansible etc.
5	Platform as a Service technology

#	Docker Components
1	Docket Client and Daemon
2	Images
3	Docker registries (Docker Hub or AWS ECR)
4	Containers

#	Containers are isolated in a host using the two Linux kernel features called **namespaces** and **control groups**
1	pid: responsible for isolating the process [PID: Process ID]
2	net: it manages the network interfaces [NET: Networking]
3	ipc: it manages access to IPC resources [IPC: Interprocess Communication]
4	mnt: responsible for managing the file system mount points [MNT: Mount]
5	uts: Isolates kernel and version identifiers [UTS: Unix Timesharing System]

Linux control groups settings decide - how much CPU and memory resource a container should use.

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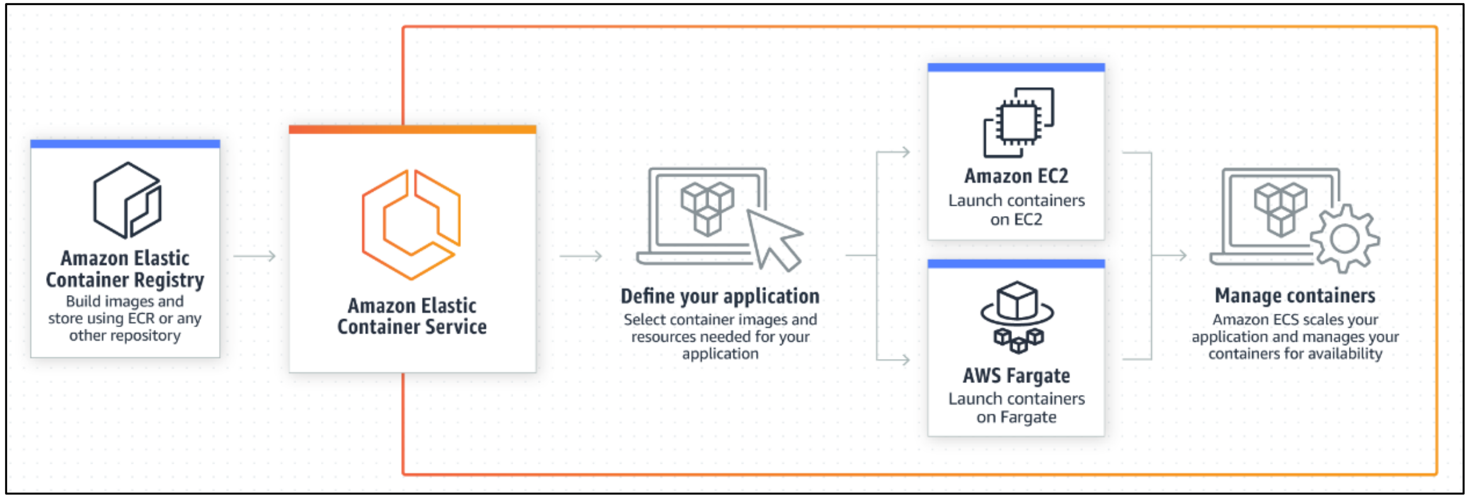
- What is Microservice
- Docker (helps CI/CD)

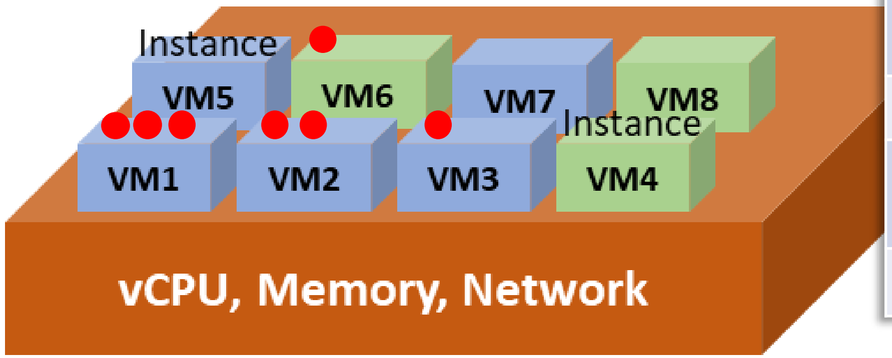
  ECS Tutorial
- Kubernetes (EKS)

  Swarm stack example



Sample docker applications

```
docker run -d -p 80:5000 training/webapp:latest python app.py
docker run -d -p 85:8000 --name whoami -t training/whoami
```



Benefits
Highly scalable, high-performance container orchestration service that supports Docker containers and allows you to easily run and scale containerized applications on AWS
Serverless processing using Fargate.
Legacy enterprise applications can be containerized and easily migrated without requiring code changes
Can be used for Batch processing and Machine Learning workloads

Learn Kubernetes at AWS EKS workshop

Serverless Vs Container Technology

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#	Topic	Serverless	Containers
1	Billing	Pay-Per-Use	Billed during idle time too
2	Scalability	Very High CSP imposed upper limit	Limitation imposed by config
3	Infrastructure Provisioning	CSP Managed	Customer Defined & Managed often cumbersome and complex setup
4	Managing Capacity	CSP Managed	Customer Defined & Managed
5	Release Downtime	No Downtime multiple versions can run simultaneously via traffic throttling	May require downtime can be automated via DevOps
6	Availability	Can work in any regional AZ	Limited to specific AZs by config
7	Invocation Latency	Must be considered for sensitive workloads	NA – as instances are continuously running and available
8	CSP Stickiness	Very High	None
9	Execution time Limit	Yes up to 15 mins max Long running apps not recommended	None
10	Processing Specs	CSP Managed	Full Control
11	Complex Architecture	Could be challenging One must consider coordination among different services	Initial setup challenges

Combination of Serverless and Container technology can bring out the best of both in an enterprise solution

Serverless - FaaS

Microservices, Docker, Kubernetes, Serverless, Service Mesh and Beyond

Choose between serverless and microservices

Cloud Adoption Framework

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Perspective	Description and Common Roles Involved
Business	Business support capabilities to optimize business value with cloud adoption. Common Roles: Business Managers; Finance Managers; Budget Owners; Strategy Stakeholders
People	People development, training, communications, and change management. Common Roles: Human Resources; Staffing; People Managers.
Governance	Managing and measuring resulting business outcomes. Common Roles: CIO; Program Managers; Project Managers; Enterprise Architects; Business Analysts; Portfolio Managers.
Platform	Develop, maintain, and optimize cloud platform solutions and services. Common Roles: CTO; IT Managers; Solution Architects.
Security	Designs and allows that the workloads deployed or developed in the cloud align to the organization's security control, resiliency, and compliance requirements. Common Roles: CISO; IT Security Managers; IT Security Analysts; Head of Audit and Compliance.
Operations	Allows system health and reliability through the move to the cloud, and delivers an agile cloud computing operation. Common Roles: IT Operations Managers; IT Support Managers.

Migration Whitepaper - Good Info to read at least once

Migration Readiness Assessment

AWS Well-Architecture Framework Tool

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- Framework
  - Designed for Cloud Architects, CXOs, Operations, Developers, etc.
  - Provides a consistent approach to evaluate architectures
  - Implement designs that will scale over time
  - Helps align specific architecture with cloud Best-practices
  - Identify areas of improvements within an architecture
  - Question set developed after working with 1000s of customer over past 10+ years by AWS team
- Understanding potential impact decisions
  - Making informed decisions about the architecture
- Uses pessimistic approach for a robust solution
  - Ask "what-if" and failure questions
  - Security related questions to reduce or mitigate risk
  - Find out alternative for a solution to improve performance at reduced cost
- Answers to many questions depends on business context
  - Nature of workload(s)
  - Size of workload(s)
  - Frequency of workload(s)…

Well-architected systems greatly increases the likelihood of business success

AWS Well-Architected Lens

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- Serverless Application
  - Best-practices for design, deploy and architect serverless application workloads
- High Performance Computing (HPC)
  - Best-practices for Design, deploy and architect High-Performance Computing (HPC) workloads
  - Placement Groups feature usage and best practices
  - Architecture best practices for
    - Loosely Coupled/High Performance Throughput Computing (aka HTC)
      - Image processing, Genomics analysis
    - Tightly Coupled/High Performance Computing
      - Computational Fluid Dynamics, Weather prediction, Reservoir simulation
  - Technology used
    - Shared Memory Parallelism (SMP)
    - Message Processing Interface (MPI)
- IoT
  - Best-practices for Design, deploy and architect IoT workloads
  - Covers
    - Device Provisioning
    - Device Telemetry
    - Device Commands &
    - Firmware Updates